What Metric Scale Can a Frozen Foundation-Model Depth Prior Give Monocular SLAM?

Abstract— **Problem. Monocular SLAM has no absolute scale of its own. A frozen FM metric-depth network looks like it could supply one, and is increasingly used as if it does. We ask the narrow, falsifiable question: *does it actually transfer metric scale, and if not, what is the smallest fix?***

What we found. On HI-SLAM2 with MapAnything over ten sequences (sixteen for the gate), it does not, and the reason is structural. The global depth rescale a metric prior would supply is an exact blind spot of the monocular factor graph (3.4×10^{-13} px reprojection change), so no reprojection, photometric, or prior-consistency residual (and no online estimator) can observe it. HI-SLAM2 realizes this gauge freedom by median-normalizing the prior at every keyframe and hard-coding the initial gauge; removing those mechanisms (a true metric-depth factor, Sec. V-H) leaves the conclusion unchanged, which is what makes the diagnosis a property of the monocular graph rather than of one codebase. No cheap image signal replaces an anchor: neither cross-view depth consistency nor a prior-free depth ratio predicts the needed correction, and an online estimator beats a fixed model on one of three sequences. Two constructive results survive. A trust gate on the *dispersion* of a short prior-free warm-up ratio flags when the network's scale is untrustworthy (AUROC 0.97 on sixteen sequences; the network's own confidence scores 0.44). And a trajectory-scale anchor recovers scale on all ten validated sequences (mean error 0.024) and thirteen of sixteen overall, where a depth-mean anchor fails (mean error 0.306). A trust-gated protocol combining the two cuts mean trajectory error from 5.9 m to 0.11 m on the ten (14.7 m to 0.33 m across all sixteen).

What to do. Treat the metric prior as a sensor to be *audited*, not trusted: gate its scale on the warm-up dispersion, and if it is trusted, anchor the gauge to the model. Otherwise spend a sparse, one-shot external metric measurement (an occasional rangefinder, stereo frame, or fiducial check) to pin trajectory scale, and use the model for dense structure only. The identifiability diagnosis and the trajectory anchor generalize across depth models; the dispersion gate is specific to MapAnything, and GT-free detection of scale uncertainty remains open.

I. Introduction

Monocular SLAM reconstructs a scene from a single moving camera. Because a photograph fixes no absolute size, the whole reconstruction is only defined up to one global factor; we call this the gauge. A reconstruction can be perfectly self-consistent and still be $3\times$ too small. Something external must fix the gauge before the output is metric.

Foundation-model monocular depth networks are an appealing external source. Many of them, including MapAnything and the metric variants of Depth Anything, predict depth in metres, and are now dropped into SLAM and Gaussian-splatting pipelines as if those metres transfer through the pipeline unchanged. This paper asks whether they do.

a) Names used throughout.: The frozen depth network is the *teacher* (MapAnything; DA-V2 Metric-Indoor and DA3METRIC for generality). The SLAM system it feeds is the *vehicle* (HI-SLAM2). The outside measurement that pins the global size is the *anchor*; we compare a *depth-mean* anchor (match the scene's average depth) with a *trajectory-scale* anchor (match a metric translation to the trajectory). r_{init} is the median ratio of teacher depth to the system's own depth over a short prior-free warm-up, and *dispersion* is its spread; AUROC scores the trust gate (0.5 = chance); ATE is trajectory error in metres after rigid alignment with scale fixed, so a scale error appears as trajectory error.

b) Setting and use case.: The camera that runs continuously is monocular, but in many robotics deployments a metric measurement is available *occasionally*: a downward rangefinder or imaging sonar [1], a stereo or RGB-D frame that is briefly mounted or momentarily reliable [2], a fiducial at a docking station [3], or GNSS in good sky [4]. Fusing episodic measurements of this kind is the classical intermittent-observation setting [5]. We take that literally: the protocol below spends roughly twenty sparse keyframe pairs *once* at startup to fix the gauge, then tracks with the monocular camera and the frozen FM prior for dense geometry. This is not continuous multi-sensor fusion; it is a bounded, one-shot metric check placed exactly where monocular SLAM is structurally blind.

c) What we did, and the short answer.: We study one concrete SLAM system (HI-SLAM2) and one frozen depth model (MapAnything) over ten sequences (sixteen for the gate), then repeat the key test with two more depth models (DA-V2 Metric-Indoor, DA3METRIC) to see what generalizes; KITTI is a documented out-of-domain control, reported as a limit of the SLAM system rather than a result about depth models. The metric scale does not transfer, and the reason is not that one system is buggy: it is an *identifiability* problem. The SLAM system realizes the monocular gauge freedom by normalizing the prior at every keyframe and fixing the initial gauge to a constant, so the model's metres are discarded before they can matter; we show the same holds after those mechanisms are removed (Sec. V-H). You must supply one external metric number. Once you do, the pipeline is metric to the accuracy of that number. The practical version is a trust gate plus a trajectory-scale anchor.

d) What we found (in order of the paper).: (i) The system discards the prior's scale by construction: swapping depth networks changes nothing (0.316 vs. 0.339), because

a hard-coded gauge constant dominates. (ii) Anchoring the gauge to the model recovers scale in-domain to the model's own bias (0.907 vs. 0.903) but transfers the wrong scale out of distribution (EuRoC 0.13–0.21). (iii) No free signal replaces the anchor, and an online estimator beats a fixed model on 1 of 3 sequences. (iv) The *spread* of the warm-up ratio is a usable trust gate *for MapAnything* (AUROC 0.97 over sixteen sequences; the model's own confidence scores 0.44). (v) The right anchor is trajectory scale, not depth mean: the latter fails (0.70/0.86/0.86 on three EuRoC sequences), the former recovers all ten (mean error 0.024) and thirteen of sixteen, failing only on difficult low-parallax sequences. (vi) A deliberate 25% local FM-depth step is absorbed by both the stock and the metric-factor vehicle: there is no live local lever. (vii) Gate + anchor cut mean trajectory error 5.9 m $\rightarrow$ 0.11 m; the diagnosis and anchor hold across three models, the specific gate does not.

e) Contributions.: (1) A measured identifiability result: the global metric-scale direction is an exact null direction of the monocular factor graph, so no reprojection, photometric, or prior-consistency residual (nor any online estimator) can recover it without an absolute metric measurement (Sec. III-B; confirmed after removing the vehicle's keyframe normalization, Sec. V-H). (2) A universal fallback: a trajectory-scale anchor that recovers scale on thirteen of sixteen sequences, where a depth-mean anchor fails (Sec. V-E). (3) A demonstrator and a fence: model-internal confidence is consistently useless for scale trust (AUROC 0.44; a constant 1.0 for DA3METRIC), while the dispersion of the initial ratio is a viable gate *for MapAnything* (AUROC 0.97); GT-free scale-uncertainty detection remains open and model-dependent (Secs. V-C–V-D).

II. Related Work

We position this as a *measured identifiability result and a protocol*, not a new SLAM back-end.

Positioning. The FM-prior SLAM line (HI-SLAM2 [6], CF2-SLAM [7], PRISM-SLAM [8], Scalix [9], VGGT-SLAM 2.0 [10]) reaches metric monocular output by *adopting* the teacher's scale, either directly or through a calibrated, probabilistic factor, and works when the teacher is in-domain. None of these can detect when adoption fails: PRISM's ray-distance factor makes the metric scale Fisher-identifiable but assumes the prior carries the right metres; Scalix treats a *learned* per-frame scale uncertainty as an independent measurement; CF2-SLAM calibrates coverage, not the scale value. Our audit supplies the missing failure statement (which direction is unobservable, measured), our **trust gate** supplies the missing failure detector (before deployment, from a prior-free reference), and our **anchor** supplies the missing fallback (pin trajectory scale when the gate rejects). The consistency accounting we propose is the missing evidence for the part of the error that *is* observable.

- **Scale-aware and metric SLAM.** Several concurrent systems add scale handling or calibration to SLAM and SLAM+GS pipelines (e.g. Scalix [9]; the FM-based line: HI-SLAM2 [6], CF2-SLAM [7], PRISM-SLAM [8], VGGT-SLAM 2.0 [10]). We do not claim those mechanisms.
- **FM depth priors.** We study metric-depth models as geometry/Gaussian priors; MapAnything [11] is our frozen teacher, with DA-V2 Metric [12] and DA3METRIC [13] for generality. Other prior sources include feed-forward geometry models [14], [15] and deep monocular odometry with a learned uncertainty [16]; our audit applies to any of them.
- **Observability.** The similarity gauge and monocular scale unobservability are classical. We give the concrete consequence for FM depth priors and a numeric certificate, plus its low-parallax face. The visual-inertial line [17], [18], [19] estimates learned-depth scale in *initialization* because the IMU makes scale observable, the external observable pure monocular SLAM lacks, and exactly what our anchor must supply.

We do not claim to introduce metric SLAM, learned covariance [20], or scale-aware factor graphs, nor online-learned model parameters for odometry [21]; this is not a bug report about HI-SLAM2: its normalization lines instantiate a gauge freedom that any anchor-free monocular pipeline has, as confirmed after they are removed (Sec. V-H).

III. How the System Handles Depth, and Why Scale is a Blind Spot

A. The setup

HI-SLAM2 is monocular factor-graph SLAM with a dense depth prior fed to a depth-only bundle adjustment at a weight of 0.01. The depth prior comes from a frozen map of depth images (one per frame) produced by the teacher, in the system's working resolution.

B. The gauge argument, and a numeric check

We state the fact we rely on once, as a proposition with a one-paragraph proof, because the corollary (not the proposition) is what the paper uses. The similarity gauge is classical; nothing below is claimed as new mathematics.

Proposition 1 (gauge null direction). *Take a pixel with depth z_i, intrinsics K, and relative pose (R, t). The reprojection residual of its backprojected point is invariant to the joint rescaling $(t, z) \rightarrow \lambda(t, z)$ for every $\lambda > 0$.*

Proof sketch. Backproject $X_i = z_i K^{-1}[u_i, 1]$ and project forward as $X_j = RX_i + t$. Under the joint rescale, $X_j \rightarrow \lambda X_j$, and projection divides out that common scale, so the image point u_j does not move, while the depth it implies grows by λ. Only the ratio t/z enters the residual. Depth-only rescaling (z alone, t fixed) is not a symmetry, because it changes that ratio. □

Corollary 1. *Any estimator state whose update direction is the joint rescale (a global depth-scale "bias" state) lies in the exact null space of the residual and is structurally unobservable. In pure monocular SLAM such a state carries no information and should not be estimated; the global metric scale can be set only by an external metric reference.* This corollary is the paper's diagnosis: it is *why* the trust gate needs

a prior-free reference and *why* an anchor is mandatory, and it bounds what any online estimator can certify. Corollary 1 is a statement about the *monocular factor graph*, not about a particular implementation: reprojection, photometric, and prior-consistency residuals are all invariant to the joint (t, z) rescale, so no arrangement of them, including a metric-depth factor at full weight, observes the scale direction. A graph with no absolute metric measurement retains this gauge freedom, and an estimate initialized in that gauge stays there.

We checked the proposition numerically. A joint (t, z) rescale changes the reprojected point by 3.4×10^{-13} px (numerical zero). A depth-only rescale (z alone, t fixed) changes it by 215.6 px, and a small rotation perturbation by 6.8 px, so the test is sensitive. The three numbers together are the certificate: the joint direction is null, the depth-only direction is not, and the measurement could have seen a difference if there were one.

C. How the shipped vehicle realizes the gauge freedom

Two mechanisms are load-bearing. First, at every keyframe the prior is median-normalized to the system's own disparity (the ratio of the two medians). Second, at warm-up the mean depth is pinned once to a constant (a scale multiplier of 2.0). Together these mean the teacher's absolute scale never reaches the estimate, and the global gauge is set once and frozen. These lines are *how this vehicle realizes* the gauge freedom, not the result: Sec. V-H removes them by wiring the prior as a true metric-depth factor, and the global conclusion is unchanged, exactly as the structural argument predicts.

IV. Method

1) **Audit.** Serve a frozen teacher and confirm the realized scale does not depend on which teacher it is (we swap Omnidata for MapAnything and get the same result).
2) **Gauge anchor.** Replace the hard-coded gauge constant with the teacher's mean metric depth. The measured relation is $\mathrm{scale}(a) = \mathrm{scale}(1)/a$, so the oracle anchor is $a = \mathrm{scale}(1)$. In-domain this makes the reconstruction metric to the teacher's own accuracy.
3) **Trust gate.** From a short *prior-free* warm-up, compute $r_{init} = \mathrm{median}(D_{teacher}/D_{backend})$ and its dispersion over the window. Trust the teacher's scale when the dispersion is low.
4) **Sparse anchor protocol (opportunistic sensor fusion).** When the gate rejects a sequence (or it is known to be out of distribution), pin metric **trajectory scale** from one short external metric translation (RGB-D or stereo PnP, matched to the backend trajectory) and use the teacher for dense structure only. A depth-mean anchor is not sufficient.

V. Experiments

a) Setup.: HI-SLAM2 (torch 2.1.2) plus frozen MapAnything depth maps over ten sequences: five from the TUM RGB-D benchmark [22] (f1_xyz, f1_desk, fr2_xyz, fr2_desk, fr3_long_office_household) and five from EuRoC [23] (MH_01, MH_02, V1_01, V1_02, V2_01). All injection is external (no edits to the SLAM code); ground truth is TUM registered depth and TUM/EuRoC trajectories.

b) Vehicle-domain control.: KITTI-800 [24] (odometry 00, first 800 frames, outdoor driving) is a documented limit of the *SLAM system*, not a result about depth models: stock HI-SLAM2 gives umeyama scale 4.74 (ATE 10.3 m) and the FM-wired run 3.86 (ATE 13.9 m). Both are wrong by roughly $4\times$ regardless of the prior, so we exclude KITTI from the prior claims.

TABLE I

Umeyama scale (est→GT) before and after anchoring the gauge to the teacher, and the teacher's own average bias; "fixed gauge" pins mean depth to 2.0 m.

sequence	fixed gauge	anchored	teacher bias
f1_xyz	0.442	**0.907**	0.903
f1_desk	0.505	**1.063**	0.907
MH_01	0.316	**0.132**	out of distribution
V1_02	0.192	**0.210**	out of distribution

TABLE II

The average warm-up ratio does not predict the needed anchor: the pairwise sign agrees three times and disagrees three times.

sequence	r_{init}	oracle anchor	domain
f1_xyz	0.478	0.907	TUM
f1_desk	0.512	1.063	TUM
MH_01	0.734	0.132	EuRoC
V1_02	0.189	0.210	EuRoC

A. The gauge dominates; anchoring recovers scale in-domain

Table I shows the effect. On TUM, anchoring recovers the teacher's own accuracy (0.907 vs. 0.903): the dominant error was the hard-coded gauge, not the network. On EuRoC the teacher's scale is itself wrong, so the same anchor transfers the wrong metres. The gauge law is linear to measurement precision: anchoring f1_xyz at 1.1 gives $0.907/1.1 = 0.825$ (measured 0.825) and f1_desk at 0.8 gives $1.063/0.8 = 1.327$ (measured 1.327). Read the adopted scale against the teacher-bias column, not against 1.0: the residual is the teacher's bias faithfully inherited.

B. No free signal replaces the anchor

We looked for a signal already in the data that would predict the correction, so an anchor would not be needed. The candidate was the warm-up ratio $r_{init} = \mathrm{median}(D_{teacher}/D_{backend})$, computed from a prior-free run at the fixed gauge. Its *average* versus the oracle anchor is in Table II.

The simple physical model ($r = \beta/\gamma$, anchor $= 1/\beta$) predicts the anchor should fall as r rises, but only if the reference gauge is the same in every domain. It is not: the pairwise signs agree three times and disagree three times, so there is no usable monotone relation, and a fit calibrated on one domain extrapolates $18\times$ wrong on MH_01.

TABLE III

SCALE ERROR OF THE FIVE-ARM LADDER. ONLINE ADAPTATION BEATS FROZEN CALIBRATION ON ONLY ONE OF THREE INFORMATIVE SEQUENCES.

sequence	fixed gauge	in-domain	frozen	online	oracle
f1_xyz	0.496	0.093	0.107	0.177	0.000
f1_desk	0.403	0.063	0.047	0.130	0.000
MH_01	0.661	0.868	0.870	0.682	0.000
V1_02	0.808	0.790	0.794	0.977	0.000

TABLE IV

TRUST LABEL IS DEFINED FROM METRIC TRUTH, NOT FROM THE DOMAIN.

label	sequences
trust (6)	f1_xyz, f1_desk, fr2_desk, fr3_long_office, V1_01, V2_01
out of dist. (4)	fr2_xyz, MH_01, MH_02, V1_02

A five-arm ladder (fixed gauge, in-domain, frozen calibration, online adaptation, oracle) tells the same story in scale error (Table III): the online arm beats the frozen arm on only one of three informative sequences. **We rejected the online-transfer route.**

C. Dispersion is a usable trust gate (for MapAnything)

The average of r_{init} does not help, but its spread does. On ten sequences, with an out-of-distribution label defined purely from metric truth ($|\text{scale} - 1| > 0.30$), the split is in Table IV.

Notice the label is not the domain: V1_01 and V2_01 are EuRoC yet trustworthy, and fr2_xyz is TUM yet out of distribution. The gate, for which low dispersion means "trust", scores **AUROC 0.917** (threshold 0.0625, accuracy 0.90), with one conservative false rejection (V2_01). The network's own confidence is useless for this: AUROC 0.44. It is confidently wrong.

The gate is not an artifact of small samples or of the warm-up window. A sequence-level bootstrap gives a mean of 0.918 with 95% CI [0.667, 1.000], and leave-one-out never drops below 0.889. Recomputing every sequence's dispersion from a *fixed* frame budget (the minimum 11 frames, versus the actual 11–106) leaves the AUROC at 0.917, and the correlation between dispersion and frame count is only -0.26. The AUC is flat across windows from 10 to 100 frames (0.833 at 5). **The gate reads the domain shift, not the sample size.**

Adding the six remaining EuRoC sequences (**16 total, 7 gate-positive**) raises the AUROC to **0.968** (bootstrap 95% CI [0.859, 1.000]; leave-one-out $\geq$ 0.963; fixed frame budget 0.968, so this is the same statistic, not a length effect). The added sequences are informative, not filler: MH_03/04/05 are correctly flagged out of distribution (dispersion 0.51/0.64/0.80) and V1_03/V2_02/V2_03 trusted (0.09/0.06/0.02). Caveat: the added reference runs use a GS-free, 300-frame vehicle configuration, so this compares the *init-window statistic* across a slightly different backend setup.

TABLE V

GATE AUROC PER DEPTH MODEL (TEN SEQUENCES). BOOTSTRAP CIS OVERLAP BECAUSE THERE ARE ONLY 4–6 OOD CASES; LEAVE-ONE-OUT SEPARATES THEM.

depth model	AUROC	95% CI	LOO
MapAnything	**0.917**	[0.667, 1.000]	0.889–1.000
DA-V2 Metric-Indoor	0.625	[0.167, 1.000]	0.500–0.778
DA3METRIC-LARGE	0.600	[0.160, 1.000]	0.500–0.700

D. Generalizes across models, but the gate is model-specific

We repeated the test with two more metric depth models. **DA-V2 Metric-Indoor-Large** [12] is less metric in many more sequences (gauge scale 0.557/0.738 on f1_xyz/f1_desk; 6 of 10 out of distribution, including two TUM sequences). **DA3METRIC-LARGE** [13] is near-metric in-domain but systematically over-estimates by 15–43% (5 of 10 out of distribution), and is badly off on EuRoC (0.23–0.26). Every model confirms the identifiability diagnosis, and every model fails differently: DA3METRIC even emits a constant confidence of 1.0, so confidence is never a scale-trust signal. The *gate*, however, does not transfer (Table V).

The point estimates separate (0.92 versus about 0.6), and the fixed-budget check gives the same number in each case, so this is not a length artifact. The bootstrap intervals overlap because there are only 4–6 out-of-distribution cases per model; the sharper statistic is leave-one-out, where both alternative models fall to 0.50 (chance) while MapAnything never drops below 0.889.

Allowing each model its own decision rule does not close the gap. Sweeping the positive-class threshold τ (defining "scale wrong" as $|\text{scale}-1| > \tau$) and calibrating τ per model, MapAnything reaches AUROC **1.00** at a stable $\tau^* = 0.20$ (leave-one-sequence-out $\tau^* = 0.215 \pm 0.045$), while DA-V2 and DA3METRIC reach only 0.875 ($\tau^* = 0.475$) and 0.810 ($\tau^* = 0.700$) at *unstable* τ^* (LOSO std 0.144 and 0.135, bootstrap CIs including 0.5). A per-model constant cannot rescue a weak ranking; the dispersion construct itself is MapAnything-specific here.

The split to carry away: **the identifiability diagnosis and the trajectory anchor are universal; the dispersion gate is a MapAnything-specific demonstration that the construct can work, not a universal detector.** Model-internal confidence is consistently uninformative (MapAnything 0.44; DA3METRIC emits a constant 1.0), and GT-free detection of scale uncertainty remains an open, model-dependent problem.

E. The right anchor is trajectory scale, not depth mean

With the gauge law, an *oracle* anchor $a = a_{gt}(1 + \epsilon)$ gives scale error $\epsilon/(1 + \epsilon)$ on every sequence (e.g. 0.000/0.091/0.167/0.231 for $\epsilon = 0.0/0.1/0.2/0.3$), independent of the scene. The question is whether a *realistic* anchor realizes it. A depth-mean anchor (match the scene's average depth) does not. Measured with real backend runs on all ten sequences, using the dataset's own sensor (TUM RGB-D or EuRoC stereo), the comparison is in Table VI.

TABLE VI

ANCHOR ERROR PER SEQUENCE FOR A DEPTH-MEAN VERSUS A TRAJECTORY-SCALE ANCHOR, WITH THE TEACHER'S OWN ERROR.

sequence	sensor (m)	teacher	depth-mean	traj.
f1_xyz	1.128	0.093	0.113	**0.009**
f1_desk	1.163	0.063	0.020	**0.069**
fr2_xyz	1.330	0.359	0.029	**0.044**
fr2_desk	1.699	0.192	0.178	**0.005**
fr3_long	2.532	0.005	0.106	**0.010**
MH_01	2.275	0.868	0.704	**0.004**
MH_02	3.248	0.903	0.856	**0.047**
V1_01	2.216	0.107	0.039	**0.011**
V1_02	2.696	0.790	0.857	**0.013**
V2_01	3.414	0.203	0.156	**0.031**
mean (10)		0.358	0.306	**0.024**

The depth-mean anchor is no better than the teacher overall (0.306 vs. 0.358), inconsistent in-domain (0.020–0.178), and catastrophic on three EuRoC sequences (0.70/0.86/0.86). The reason is the one we found earlier: the average depth that makes the *sensor* metric and the average depth that makes the *trajectory* metric are different, badly so out of distribution. A trajectory-scale anchor, a metric translation matched to the backend trajectory, recovers all ten sequences (mean **0.024**), using 11–27 matched pairs each with a bootstrap interval. Dense short-baseline pairs across the trajectory also remove our earlier failure cases (f1_desk is now 0.069, not degenerate; V1_02 rests on 12 pairs, not 1).

Extending the anchor to the six remaining EuRoC sequences tests where it stops working. It accepts **13 of the 16** sequences (mean error **0.021**, worst 0.069); the three it rejects are MH_04, MH_05 and V2_03, the hardest sequences, where matched stereo pairs are poorest and the pair-ratio interval widest (V2_03 fails on a CI half-width above 0.25, the other two on scale). The three new sequences it does recover have teacher errors of 0.93/0.03/0.11 and anchor errors of 0.003/0.010/0.010. This is the anchor's practical low-parallax limit (Sec. V-F): it is exact where matched translation exists and refuses where it does not.

F. The low-parallax limit, stated exactly

The trajectory anchor needs translation, and the underlying limit is exact. Holding the pose fixed, a 10% depth-scale error changes the reprojection by 2.3×10^{-13} px at zero translation (depth is a pure gauge there), and the change grows monotonically with baseline: 0.007 px at $|t| = 10^{-4}$, 0.74 at 0.01, 7.0 at 0.1, 42 at 1.0. Real keyframe baselines are far from this limit: the anchor's scale stays within a few percent across baseline terciles, with mean interval half-width 0.147 (low) versus 0.133 (high). **So the teacher-only failure is a fact about identifiability, not a side effect of low parallax.**

G. The sparse-anchor protocol, measured in metres

Composing the gate (threshold 0.0625) with the trajectory anchor gives the paper's practical result, measured as rigid-aligned trajectory error so that a scale error shows up as metric error:

- **Teacher scale only (the naive default):** mean scale error 0.358, mean ATE **5.94 m**.
- **Gate + trajectory anchor:** mean scale error 0.060, mean ATE **0.11 m**.

The right way to read this is **catastrophe prevention, not a marginal ATE improvement**. "Teacher scale only" is not a weak baseline: it is the default of deploying an FM metric-depth prior and trusting its metres everywhere. On the four out-of-distribution sequences it produces trajectory errors of tens of metres. The protocol does not make an already-good system slightly better; it turns a silent, unbounded failure into a bounded one.

Five sequences are trusted and use the teacher's scale directly (mean error 0.092). The other five are rejected and use the trajectory anchor; teacher-only would have erred 0.625 on them (0.730 on the four truly out-of-distribution cases). The four worst cases are all recovered: MH_01 $30.0 \rightarrow 0.05$ m, MH_02 $21.4 \rightarrow 0.11$ m, V1_02 $6.7 \rightarrow 0.07$ m, fr2_xyz $0.14 \rightarrow 0.01$ m. The gate is conservative but safe, and the anchor makes a false rejection cost nothing.

Applied across all sixteen sequences, including the three where the anchor cannot be validated, the gate+anchor protocol still cuts mean rigid ATE from **14.7 m to 0.33 m** (scale error 0.417→0.083; the sixteen-sequence gate uses its own threshold 0.0979, the ten uses 0.0625). The residual is concentrated entirely in the difficult cases: MH_04 $66.7 \rightarrow 2.0$ m and MH_05 $59.5 \rightarrow 1.5$ m, where the anchor itself fails, while the third anchor failure, V2_03, is trusted by the gate and stays at its teacher error (0.36 m). The ten-sequence result is the clean statement; the sixteen-sequence run shows the protocol degrades boundedly rather than silently. Figs. 1 and 2 summarize the gate and protocol. Table VII lists the per-sequence outcome; the protocol spends 264 sparse pairs in total (11–27 per sequence).

What the protocol costs. The anchor needs only 11–27 sparse keyframe pairs, applied *once* at startup, and the gate needs only a short prior-free warm-up: roughly twenty measurements to bound the monocular failure, after which tracking is pure monocular with the FM prior supplying dense geometry. The metric source can be opportunistic (a rangefinder or sonar, a brief stereo or RGB-D frame, a docking fiducial, GNSS) and is not committed to the front end [5], [1], [2], [3], [4]. The one case this does not help is a platform whose *only* sensor is RGB-D; then monocular SLAM has no advantage and the protocol is moot.

H. The global gauge has no local lever

If the global gauge is a blind spot, perhaps local model scale is trackable. It is not, in either vehicle. In the shipped system a deliberate 25% step in the model's depth from frame 400 leaves the global scale unmoved (0.8246 vs. 0.8252) and the per-keyframe depth pass-through at the noise floor ($\sim$0.0016), because of the per-keyframe renormalization. Wired as a true **metric-depth factor** (scale grid removed, prior entering at its own metres), the global result is unchanged: the scalar and confidence-weighted arms sit at or slightly worse than stock (mean $|\text{scale}-1| = 0.523$ vs. 0.409) and never approach

TABLE VII

PER-SEQUENCE OUTCOME OF THE GATE+ANCHOR PROTOCOL OVER ALL SIXTEEN SEQUENCES. "DISP." IS THE WARM-UP DISPERSION $\sigma_{\log}$ (THRESHOLD 0.0979), "SRC" THE SCALE SOURCE FINALLY USED, AND ATE IS RIGID-ALIGNED TRAJECTORY ERROR IN METRES. THE EIGHT TRUSTED SEQUENCES KEEP THE TEACHER'S SCALE; THE EIGHT REJECTED ONES TAKE THE TRAJECTORY ANCHOR. THE TWO WORST REJECTIONS (MH_04, MH_05) ARE THE ANCHOR'S OWN FAILURES.

sequence	disp.	gate	src	pairs	ATE teacher (m)	ATE protocol (m)
f1_xyz	0.012	trust	FM	23	0.020	**0.020**
f1_desk	0.022	trust	FM	13	0.055	**0.055**
fr2_xyz	0.474	reject	anchor	18	0.138	**0.012**
fr2_desk	0.061	trust	FM	11	0.464	**0.464**
fr3_long	0.010	trust	FM	13	0.016	**0.016**
MH_01	0.240	reject	anchor	27	30.029	**0.050**
MH_02	0.288	reject	anchor	17	21.429	**0.109**
MH_03	0.510	reject	anchor	19	49.454	**0.046**
MH_04	0.642	reject	anchor	25	66.731	**1.994**
MH_05	0.802	reject	anchor	17	59.457	**1.511**
V1_01	0.020	trust	FM	11	0.173	**0.173**
V1_02	0.621	reject	anchor	12	6.664	**0.073**
V1_03	0.094	trust	FM	15	0.084	**0.084**
V2_01	0.382	reject	anchor	13	0.369	**0.090**
V2_02	0.059	trust	FM	15	0.265	**0.265**
V2_03	0.022	trust	FM	15	0.361	**0.361**
mean				16.5	14.732	**0.333**

TABLE VIII

UMEYAMA SCALE (EST→GT) PER METRIC-FACTOR ARM. THE METRIC RESIDUAL DOES NOT MAKE THE GLOBAL GAUGE OBSERVABLE: THE SCALAR AND CONFIDENCE-WEIGHTED ARMS NEVER APPROACH THE ANCHORED RESULT.

sequence	stock	scalar	conf.	metric+anch.	anchor only
f1_xyz	0.505	0.505	0.499	0.896	0.907
f1_desk	0.597	0.577	0.538	0.964	1.063
fr2_xyz	0.680	0.748	0.750	0.708	0.641
fr2_desk	1.000	1.088	1.090	0.823	0.808
fr3_long	1.132	1.450	1.432	1.288	1.005
MH_01	0.339	0.270	0.353	0.133	0.132
MH_02	0.231	0.271	0.265	0.125	0.097
V1_01	1.065	1.333	1.351	1.403	1.107
V1_02	0.192	0.212	0.242	0.036	0.210
V2_01	1.439	1.881	2.008	1.607	1.203
mean	0.409	0.517	0.523	0.462	0.358

the anchored result; raising the prior weight α from 0.01 to 10 reproduces the $0.44 \to 0.50$ shift but never reaches the teacher's 0.90 (Table VIII). **The failure survives removing the per-keyframe normalization entirely, so it is the gauge null of the monocular graph, not HI-SLAM2's implementation.** A deliberate local step is likewise absorbed by both vehicles, and per-pixel confidence weighting changes shape, not scale.

I. The observable half: a scalar calibrates; a per-regime extension is fragile

A scalar variance scale on the observable half can be calibrated online. We weight the metric-factor residual by the teacher's confidence and adapt a variance scale online by covariance matching [25], reporting the depth-channel NIS (target 1; we use the median over solves because rare diverging solves dominate the mean, on MH_01 a few solves reach NIS $> 10^5$). Frozen confidence weighting is miscalibrated by a sequence-dependent factor: median NIS 0.477 (f1_xyz), 0.074 (f1_desk), 0.021 (MH_01), 0.082 (V1_02), so no single temperature covers all. A scalar adaptation brings every sequence closer to 1 (1.380/0.469/0.033/0.025), closer than frozen on all four, but on the extreme-OOD MH_01 it pins its floor ($c = c_{\min} = 0.2$) at NIS 0.033, because the miscalibration is not a global temperature.

Replacing the scalar with K=3 confidence-quantile regimes (edges calibrated from the first 60 solves, then edges and c_k frozen at 300 solves, held-out on later tracking solves) passes the pre-registered test in this one configuration (Table IX): per-regime σ is closer to 1 than the scalar on all three shifted sequences and reduces the mean per-regime deviation $|\mathrm{NIS}_k - 1|$ on all four. **It is not robust.** The pre-registered K-sensitivity ($K \in \{2, 4\}$) fails: unguarded, K2/K4 hung in the backend on both shifted sequences, and the surviving K3 pins the guard floor ($c_k = 0.05$) and diverges on 11% of held-out MH_01 solves. Re-running K2/K3/K4 under a stability guard ($c_{lo} = 0.2$, time-capped) makes every arm converge but *erases the calibration gain* (MH_01 K3 median 0.032 and V1_02 K3 median 0.019, versus 0.504/0.615 unguarded). So per-regime σ is a single fragile configuration, not a mechanism; only the scalar adaptation is robust. The calibration is also **domain-specific**: freezing the TUM-fitted scalar and applying it to EuRoC leaves held-out NIS at 0.03–0.09, far from 1, so the observable half must be re-calibrated per domain. Depth-channel NIS is circular for absolute scale by construction.

Finally, the gate is **blind to a constant teacher bias**. Rebuilding the teacher with perturbed intrinsics ($fx, fy \times 1.20$) induces a constant ∼5% teacher depth bias (the model partly compensates the focal change). The gate's dispersion barely moves and stays below threshold for the trusted sequences, while the anchored scale shifts with the bias (Table X). This is by construction: r_{init} is a ratio against the backend, so a

TABLE IX

SCALAR VERSUS $K{=}3$ PER-REGIME ADAPTATION: DEPTH-CHANNEL MEDIAN NIS (TARGET 1) AND PER-REGIME NIS. PER-REGIME IS CLOSER TO 1 BUT NOT ROBUST TO K.

sequence	scalar med.	regime med.	scalar NIS$_k$	regime NIS$_k$
f1_xyz	1.380	1.315	[0.60,2.00,2.78]	[1.33,1.34,1.24]
f1_desk	0.469	0.707	[0.12,0.37,1.05]	[0.50,0.36,0.67]
MH_01	0.033	0.504	[0.01,0.01,0.04]	[0.04,0.07,0.67]
V1_02	0.025	0.615	[0.01,0.01,0.03]	[0.16,0.17,0.64]

TABLE X

THE GATE IS BLIND TO A CONSTANT TEACHER BIAS: PERTURBING THE INTRINSICS SHIFTS THE ADOPTED SCALE BUT BARELY MOVES THE GATE STATISTIC.

		dispersion		adopted scale	
sequence	gate	real	pert.	real	pert.
f1_xyz	trust	0.012	0.019	0.907	0.877
f1_desk	trust	0.022	0.026	1.063	1.022
MH_01	reject	0.240	0.260	0.132	0.120

constant multiplicative teacher bias rescales it uniformly and leaves its dispersion, the gate statistic, unchanged. **The gate detects teacher inconsistency/OOD, not a globally biased teacher; a constant bias must be caught by the anchor.** We state this as a designed limitation. Fig. 3 collects these five supporting results.

VI. DISCUSSION AND LIMITATIONS

a) Why the free-signal route fails.: The reference used by r_{init} is itself only up-to-scale, so the relation between r and the needed correction requires a domain-independent reference gauge that monocular geometry does not provide. This is the observability statement, not a bug in our implementation.

b) Two halves of the error; scale is not adaptive.: Corollary 1 splits the prior's error into a part the residuals can see (per-pixel σ, the relative scale across keyframes, the shape) and a part they cannot: the single global scale direction. A scalar part of the observable half is a legitimate target for online estimation and consistency checking (Sec. V-I); the unobservable half cannot be estimated at all and must instead be *gated* and *anchored* by an external reference. This is why the two constructive pieces do not overlap, and why the honest framing for the global direction is one-shot identifiability plus an anchor, not adaptive recovery of scale. Depth-channel consistency checks are *circular* for absolute scale (the prior defines the scale they would validate), and image-channel residuals are blind to the joint rescale (Proposition 1): an adaptive estimator can certify only what the gauge lets it see.

c) Practical protocol.: (1) Gate the model's metric scale on the dispersion of the prior-free warm-up ratio. (2) If trusted, anchor the gauge to the model. (3) Otherwise pin trajectory scale with a metric translation and use the model for dense structure only; a depth-mean anchor is not sufficient.

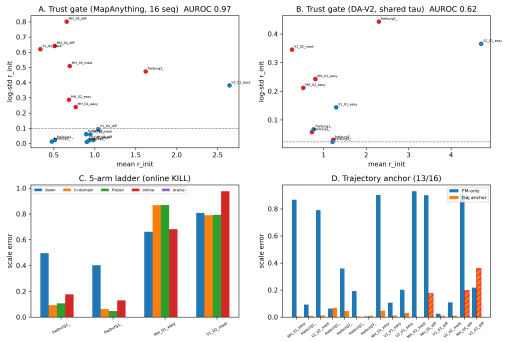


Fig. 1. (A) trust gate, MapAnything, sixteen sequences (AUROC 0.97); (B) trust gate, DA-V2 Metric at a shared threshold (AUROC 0.62; the gate is model-specific); (C) the five-arm ladder (the online route we rejected); (D) trajectory anchor versus teacher-only across sixteen sequences; the hatched bars are the three sequences the anchor rejects.

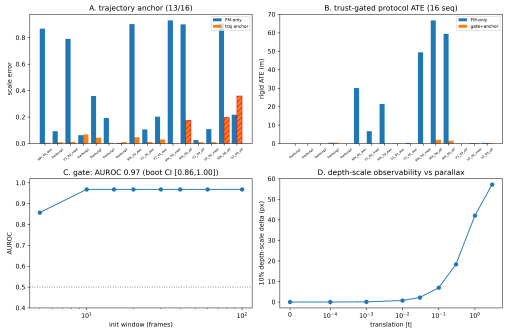


Fig. 2. (A) trajectory anchor versus teacher-only, sixteen sequences (13 accepted); (B) trust-gated protocol error across sixteen sequences, teacher-only versus gate+anchor; (C) gate init-window sweep with bootstrap interval (sixteen sequences); (D) depth-scale visibility versus parallax.

d) What generalizes, and what does not.: The gauge diagnosis (Corollary 1) and the trajectory anchor hold across the three depth models and both in- and out-of-distribution domains; these are the transferable results. The trust gate does not: the dispersion of the initial ratio separates scale errors for MapAnything (AUROC 0.97) but not for DA-V2 (0.62) or DA3METRIC (0.60), and model-internal confidence is useless everywhere we looked (0.44, or a constant 1.0). General GT-free detection of metric-scale uncertainty is therefore an open problem, not a solved one.

e) What would falsify this.: The diagnosis fails if a purely monocular residual changes under a joint (t, z) rescale; none does, and the numeric certificate checks the reprojection term. The anchor's universality fails if a depth-mean or model-internal anchor matches its accuracy across domains; on our ten sequences it does not (0.306 and 0.358 versus 0.024). The gate's model-specificity fails if some model's dispersion ranks scale error well across domains; for DA-V2 and DA3METRIC it does not (AUROC 0.62 and 0.60).

f) Limitations.: Sixteen sequences (five TUM, eleven EuRoC), three metric models, one SLAM system. The six added EuRoC reference runs use a GS-free, 300-frame vehicle configuration (the Gaussian backend crashes on some of those visuals), so that part of the gate comparison is across a slightly different backend setup. The gate is model-specific and must

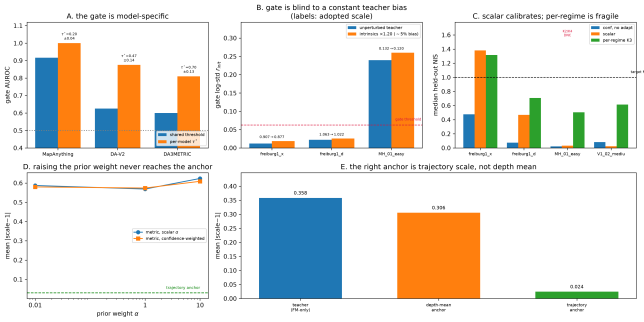


Fig. 3. The five results stated above, shown together: (A) the gate is model-specific: a per-model threshold lifts DA-V2/DA3METRIC to 0.88/0.81 but at unstable τ^* (LOSO std 0.14/0.13), while MapAnything reaches 1.00 at a stable $\tau^* = 0.20$; (B) the gate is blind to a constant teacher bias; (C) scalar adaptation reaches NIS ≈ 1 while the per-regime extension is fragile (K2/K4 did not converge unguarded); (D) raising the prior weight α through three orders of magnitude leaves the metric-factor arms near mean $|\text{scale} - 1| \approx 0.6$, far from the trajectory anchor's 0.03; (E) of the three anchors only the trajectory one works: teacher-only 0.358, depth-mean 0.306, trajectory 0.024 mean $|\text{scale} - 1|$ over ten sequences.

be validated per model; it detects teacher inconsistency, not a constant bias, which by construction leaves the gate statistic unchanged (Sec. V-I), so a globally biased teacher must be caught by the anchor. Local-drift testing is on TUM only (EuRoC has no dense ground-truth depth); the trajectory anchor still needs an external metric sensor, and it recovers 13 of the 16 sequences, refusing the three difficult EuRoC sequences where matched translation is too poor to certify a scale; the sixteen-sequence gate and protocol figures include those failures rather than excluding them. KITTI is reported as a limit of the SLAM system, not a result about depth models. The per-regime σ of Sec. V-I passes the pre-registered test in one configuration but is **not robust**: K2/K4 hung unguarded and a stability guard erased the calibration gain, so only the scalar adaptation is reported as a mechanism. A deliberate local FM step is likewise absorbed by the metric-factor vehicle (Sec. V-H), so we report no live local lever.

VII. Conclusion

A frozen FM metric-depth prior does not transfer metric scale to monocular SLAM on its own. The monocular gauge makes the *global* scale direction unobservable (Corollary 1), and the system as shipped normalizes away even local model scale per keyframe. The diagnosis is implementation-independent: it is confirmed after those normalizations are removed (Sec. V-H), so it is a property of the monocular factor graph, not of one codebase. For the global direction the fix is one external metric anchor plus a GT-free trust gate, and no cleverer bias or covariance estimator can substitute. This is a statement about scale, not about adaptation in general: a scalar variance scale on the observable half can be calibrated online, though a per-regime refinement was not robust, provided neither is mistaken for scale recovery. We give the measured chain and a minimal protocol.

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